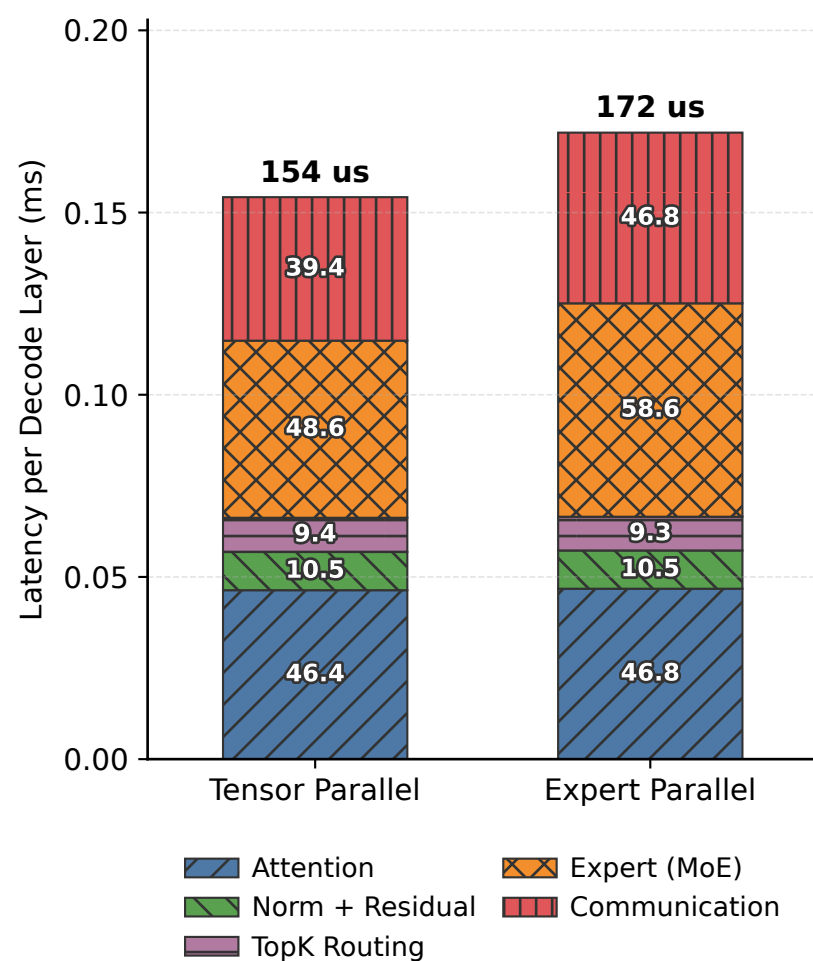




Tensor Parallel

Attention

```
flash_fwd_splitkv_combine_kernel<Flash_fwd_kernel_traits<(const int)5, (Flash_fwd_kernel_traits>, (int)4, (int)5, (bool)1>  
flash_fwd_splitkv_kernel<Flash_fwd_kernel_traits<(const int)5, (Flash_fwd_kernel_traits>, (int)4, (int)5, (bool)1>  
kernel<int>, cublasGemmTensorStridedBatched<__nv_bfloat16>, float>>  
reshape_and_cache_flash_kernel<__nv_bfloat16>  
triton_poi_fused_view_5  
triton_poi_fused_view_6
```

Norm + Residual

```
triton_poi_fused_add_mul_sub_3  
triton_poi_fused_add_mul_sub_4  
triton_poi_fused_add_mul_sub_5  
triton_red_fused_to_copy_add_mean_mul_pow_rsqrt_0  
triton_red_fused_to_copy_add_mean_mul_pow_rsqrt_1  
triton_red_fused_to_copy_mean_pow_2  
triton_red_fused_to_copy_mean_pow_3
```

TopK Routing

```
[lambda(float, float) (instance 1)]>, unsigned int, float, (int)4, (int)4>>  
[lambda(int) (instance 1)]>  
topkGatingSoftmax<(int)4>
```

Expert (MoE)

```
BFloat16, (int)4, (int)4>>  
BFloat16>, (bool)1>  
StoreWithCast<(int)1>>  
ampere_bf16_s16816gemm_bf16_64x64_sliced1x2_ldg8_f2f_stages_64x6_tn  
count_and_sort_expert_tokens_kernel<int>  
fused_moe_kernel  
moe_align_block_size_kernel<int>
```

Communication

```
ncclDevKernel_AllGather_RING_LL  
ncclDevKernel_Reduce_Sum_bf16_RING_LL
```

Expert Parallel

Attention

```
flash_fwd_splitkv_combine_kernel<Flash_fwd_kernel_traits<(const int)5, (Flash_fwd_kernel_traits>, (int)4, (int)5, (bool)1>  
flash_fwd_splitkv_kernel<Flash_fwd_kernel_traits<(const int)5, (Flash_fwd_kernel_traits>, (int)4, (int)5, (bool)1>  
kernel<int>, cublasGemmTensorStridedBatched<__nv_bfloat16>, float>>  
reshape_and_cache_flash_kernel<__nv_bfloat16>  
triton_poi_fused_view_5  
triton_poi_fused_view_6
```

Norm + Residual

```
triton_poi_fused_add_mul_sub_3  
triton_poi_fused_add_mul_sub_4  
triton_poi_fused_add_mul_sub_5  
triton_red_fused_to_copy_add_mean_mul_pow_rsqrt_0  
triton_red_fused_to_copy_add_mean_mul_pow_rsqrt_1  
triton_red_fused_to_copy_mean_pow_2  
triton_red_fused_to_copy_mean_pow_3
```

TopK Routing

```
[lambda(float, float) (instance 1)]>, unsigned int, float, (int)4, (int)4>>  
[lambda(int) (instance 1)]>  
topkGatingSoftmax<(int)4>
```

Expert (MoE)

```
BFloat16, (int)4, (int)4>>  
BFloat16>, (bool)1>  
StoreWithCast<(int)1>>  
ampere_bf16_s16816gemm_bf16_64x64_sliced1x2_ldg8_f2f_stages_64x6_tn  
count_and_sort_expert_tokens_kernel<int>  
fused_moe_kernel  
moe_align_block_size_kernel<int>  
triton_poi_fused_2  
triton_poi_fused_to_copy_gather_index_ones_like_scatter_add_1  
triton_poi_fused_to_copy_ones_like_scatter_add_0
```

Communication

```
ncclDevKernel_AllGather_RING_LL  
ncclDevKernel_Reduce_Sum_bf16_RING_LL
```